Machine Learning for Biology and Health Homework 3 Writeup

Data and Preprocessing:

Heart Disease Model:
The heart disease dataset had no columns with missing data. I scaled the model using StandardScaler and then used PCA after all in a pipeline in order to prevent data leakage. I kept all of the features in the X set I used to train and test the model.

```
               age          sex           cp     trestbps        chol  \
count  1025.000000  1025.000000  1025.000000  1025.000000  1025.00000
mean     54.434146     0.695610     0.942439   131.611707   246.00000
std       9.072290     0.460373     1.029641    17.516718    51.59251
min      29.000000     0.000000     0.000000    94.000000   126.00000
25%      48.000000     0.000000     0.000000   120.000000   211.00000
50%      56.000000     1.000000     1.000000   130.000000   240.00000
75%      61.000000     1.000000     2.000000   140.000000   275.00000
max      77.000000     1.000000     3.000000   200.000000   564.00000

               fbs      restecg      thalach        exang      oldpeak  \
count  1025.000000  1025.000000  1025.000000  1025.000000  1025.000000
mean      0.149268     0.529756   149.114146     0.336585     1.071512
std       0.356527     0.527878    23.005724     0.472772     1.175053
min       0.000000     0.000000    71.000000     0.000000     0.000000
25%       0.000000     0.000000   132.000000     0.000000     0.000000
50%       0.000000     1.000000   152.000000     0.000000     0.800000
75%       0.000000     1.000000   166.000000     1.000000     1.800000
max       1.000000     2.000000   202.000000     1.000000     6.200000
```

Biological Aging Model:
The Biological Aging had several gene expression marker columns with missing entries. I again scaled the model using StandardScaler and then used PCA in a pipeline. For the X set, I dropped the dataset, tissue_type, and gender features primarily due to the values being categorical and not float or integer based. Additionally, I dropped the age feature as that is the column we are considering our result we are trying to predict.

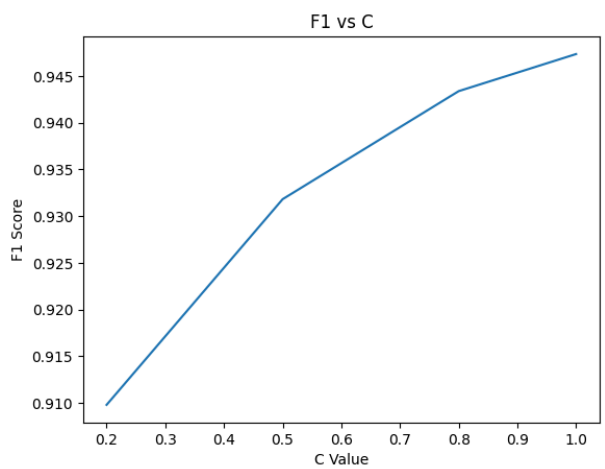
```
             age  cg00000029  cg00000108  cg00000109
count  37.00000   37.000000   37.000000   37.000000
mean   73.00000    0.079491    0.921897    0.897099
std     7.94425    0.024670    0.011144    0.019253
min    65.00000    0.049709    0.898088    0.856851
25%    67.00000    0.064024    0.914084    0.886380
50%    69.00000    0.077965    0.921435    0.898524
75%    80.00000    0.084811    0.927894    0.906837
max    94.00000    0.182037    0.946394    0.940728

        cg00000289  cg00000292  cg00000321  cg00000363
count   37.000000   37.000000   37.000000   37.000000
mean     0.184302    0.118944    0.103223    0.092733
std      0.035983    0.041468    0.016018    0.016908
min      0.113309    0.069737    0.072656    0.069040
25%      0.155735    0.087095    0.094618    0.084008
50%      0.184330    0.107637    0.101860    0.089881
75%      0.206587    0.134657    0.111954    0.095879
max      0.276800    0.229434    0.145136    0.161318
```

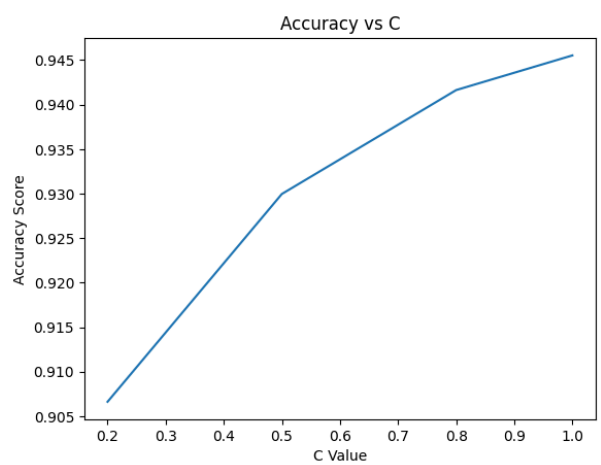
Modeling and Hyperparameters:

Heart Disease Model:

For the heart disease model, the poly kernel worked the best, having the best accuracy and recall scores by far and also having the best F1 score.

```
LINEAR
{'Accuracy': 0.8521400778210116, 'Precision': 0.8027210884353742, 'Recall': 0.9291338582677166, 'F1': 0.8613138686131386}

POLY
{'Accuracy': 0.9416342412451362, 'Precision': 0.9057971014492754, 'Recall': 0.984251968503937, 'F1': 0.9433962264150944}

RBF
{'Accuracy': 0.9377431906614786, 'Precision': 0.9172932330827067, 'Recall': 0.9606299212598425, 'F1': 0.9384615384615385}
```

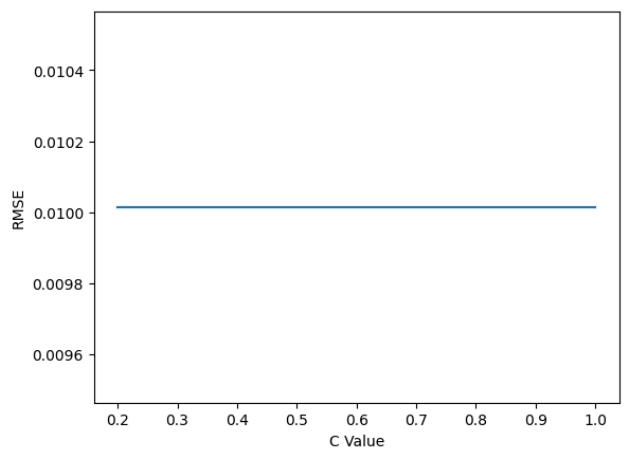




Additionally, higher c values directly translated to better F1 scores (closer to 1) and better accuracy values (closer to 1).

Biological Aging Model:
The linear kernel worked much better for prediction on the biological aging dataset. It was the only kernel that resulted in a positive R2 value and had a much lower mean squared error and mean absolute error than the other two.

```
LINEAR
{'Mean Absolute Error': 3.8262787033557886, 'Mean Squared Error': 18.972711216294538, 'R2': 0.5562976796937666}

POLY
{'Mean Absolute Error': 6.0777744432602985, 'Mean Squared Error': 43.58510309024293, 'R2': -0.019296143363960017}

RBF
{'Mean Absolute Error': 4.440000000000003, 'Mean Squared Error': 45.31999999999997, 'R2': -0.05986903648269326}
```



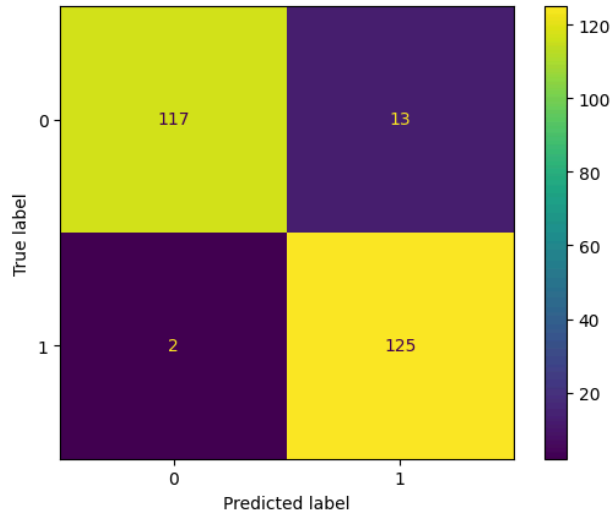
Additionally, changes in the C hyperparameter did not cause any change in the mean squared error.

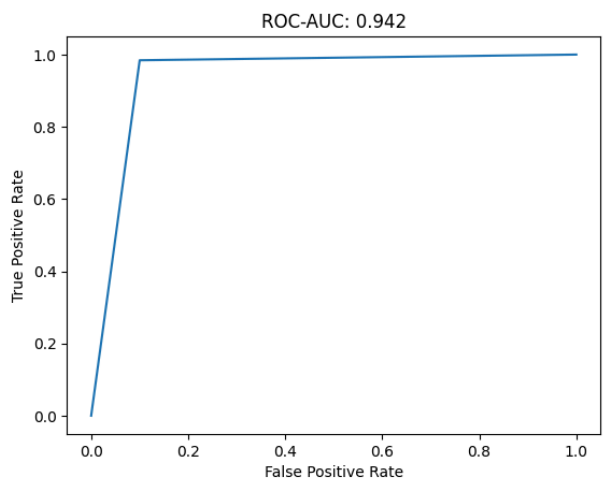
It makes sense that the classification problem worked best with the poly kernel as it does not have a simple linearly separable relationship. Instead the poly kernel was able to learn more complex relationships that more accurately mapped the classification predictions. On the other hand, the regression problem is much more likely to follow a somewhat straight line path of features increases to prediction increases, so it makes sense the linear kernel worked well for it.

Metrics and Visuals:
Heart Disease Model:

The final heart disease SVM model using the poly kernel achieved an accuracy of 94%, precision of 90.5%, recall of 98.4%, and F1 score of 94.3%. The model primarily suffered with false positives (13 out of 117) and was very successful at predicting ground truth true labels (mislabeling only 2 out of 127). Additionally, it achieved an area under the ROC curve of 0.942.

```
POLY
{'Accuracy': 0.9416342412451362, 'Precision': 0.9057971014492754, 'Recall': 0.984251968503937, 'F1': 0.9433962264150944}
```

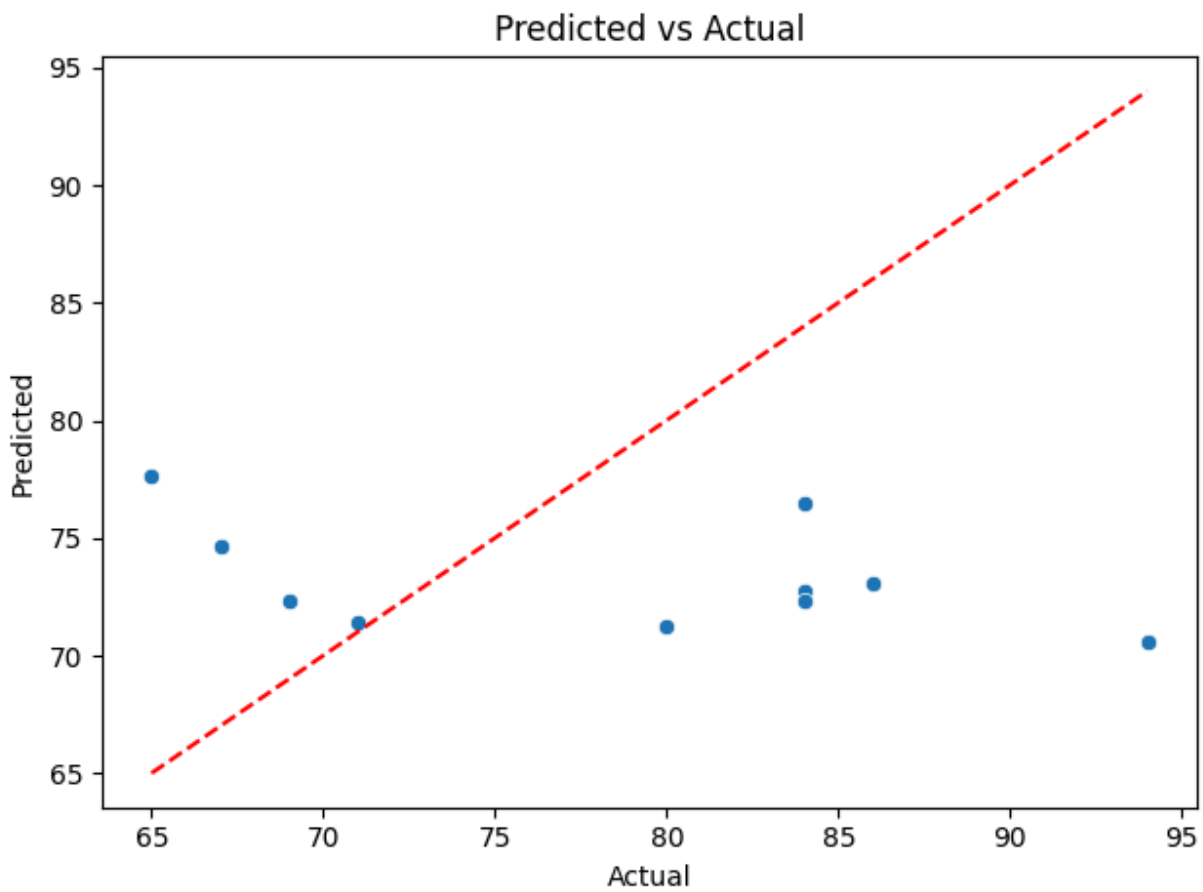




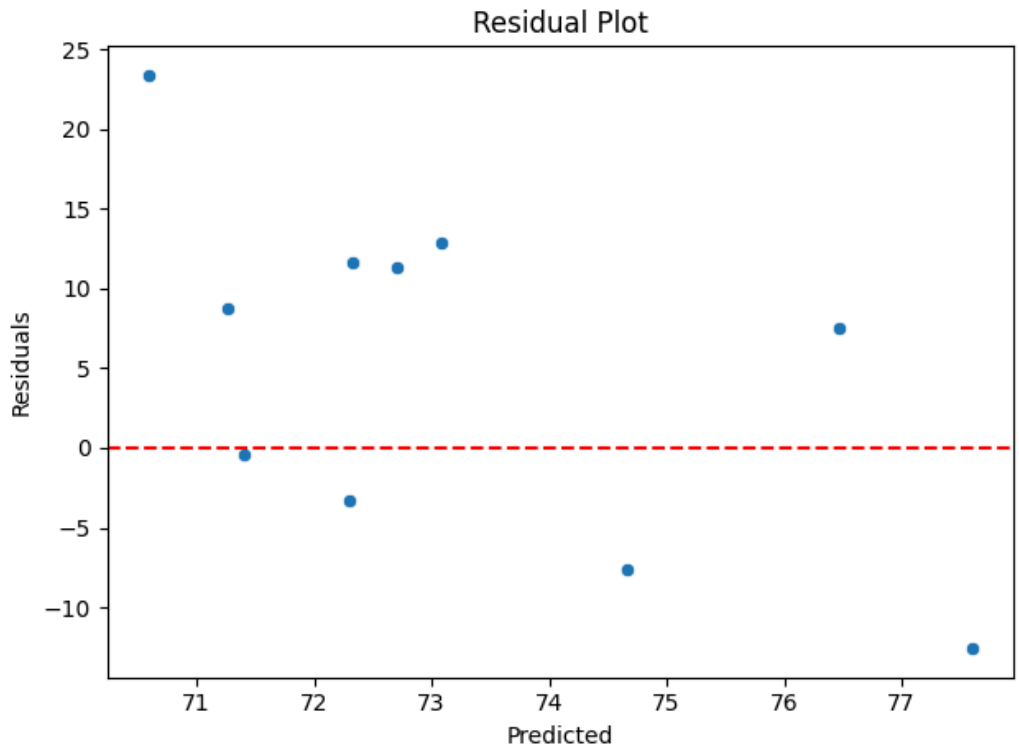
Biological Aging Model:

The final biological model using the linear kernel achieved a mean absolute error of 3.83, mean squared error 18.97, and R2 score of 0.56. These scores were very good relative to the other two kernels (as shown in the hyperparameter tuning section).

```
LINEAR
{'Mean Absolute Error': 3.8262787033557886, 'Mean Squared Error': 18.972711216294538, 'R2': 0.5562976796937666}
```



The predicted vs actual plot unfortunately does not show a very significant relationship. The values are very spread out and do not generally follow the ideal line which is marked in red. This is most likely due to the very high dimensionality and amount of noise in the dataset.



Similarly the residual plot shows several points with significant errors. This again can likely be attributed to the large amount of noise and high dimensionality of the dataset.

Sensitivity:
Heart Disease Model:
Before PCA:

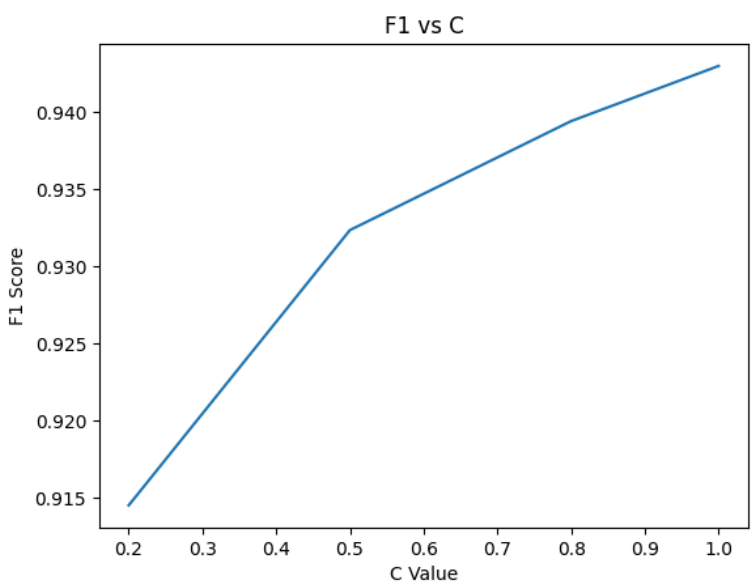
```
LINEAR
{'Accuracy': 0.8210116731517509, 'Precision': 0.8071428571428572, 'Recall': 0.8560606060606061, 'F1': 0.8308823529411765}

POLY
{'Accuracy': 0.9221789883268483, 'Precision': 0.9375, 'Recall': 0.9090909090909091, 'F1': 0.9230769230769231}

RBF
{'Accuracy': 0.9066147859922179, 'Precision': 0.9153846153846154, 'Recall': 0.9015151515151515, 'F1': 0.9083969465648855}
```

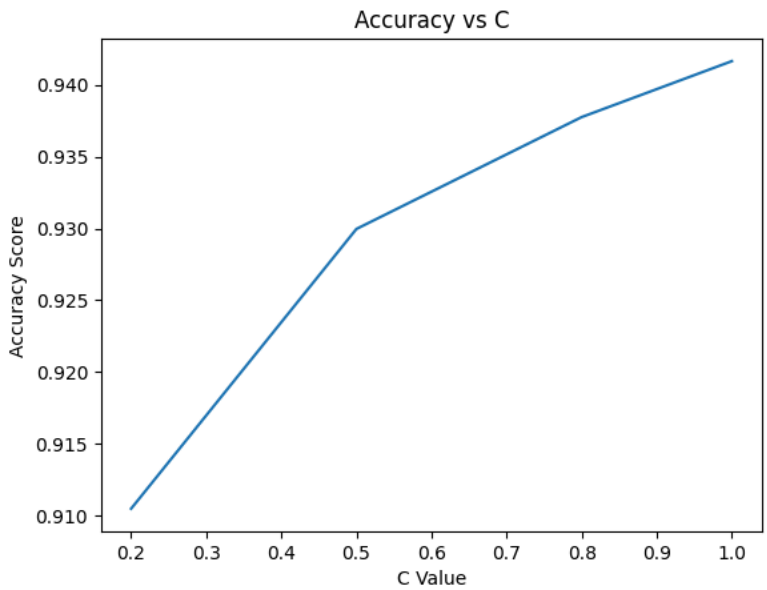
After PCA:

```
LINEAR
{'Accuracy': 0.8404669260700389, 'Precision': 0.7785234899328859, 'Recall': 0.9354838709677419, 'F1': 0.8498168498168498}

POLY
{'Accuracy': 0.9494163424124513, 'Precision': 0.9302325581395349, 'Recall': 0.967741935483871, 'F1': 0.9486166007905138}

RBF
{'Accuracy': 0.9494163424124513, 'Precision': 0.9302325581395349, 'Recall': 0.967741935483871, 'F1': 0.9486166007905138}
```

The heart disease model performed better after adding principal component analysis. This might be due to the model having less features with less noise and significantly more variance after the pca process. This process might have also helped prevent overfitting.

Testing C Values:





As stated earlier, higher c values directly translated to better F1 scores and better accuracy values.

Biological Aging Model:
Before PCA:

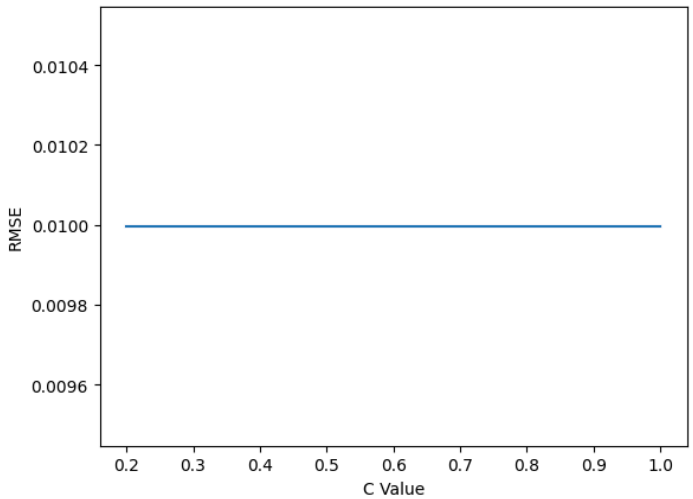
```
LINEAR
{'Mean Absolute Error': 4.853024885855052, 'Mean Squared Error': 31.317792614597373, 'R2': 0.3487670489790523}

POLY
{'Mean Absolute Error': 5.760892882371195, 'Mean Squared Error': 45.49041467174855, 'R2': 0.05405667141300563}

RBF
{'Mean Absolute Error': 5.395000000000001, 'Mean Squared Error': 50.730624999999975, 'R2': -0.054910064462465824}
```

After PCA:

```
LINEAR
{'Mean Absolute Error': 5.100928557460105, 'Mean Squared Error': 41.38143407505926, 'R2': 0.04231811906828831}

POLY
{'Mean Absolute Error': 5.200568001668953, 'Mean Squared Error': 41.26270349726867, 'R2': 0.045065876017850726}

RBF
{'Mean Absolute Error': 4.75333333333333, 'Mean Squared Error': 49.13111111111119, 'R2': -0.13703103705418163}
```

Interestingly, the biological aging model performed worse after adding principal component analysis(pca) into the pipeline. This might be due to pca getting rid of features which contained important predictive information.

Testing C Values:



Changes in the C value of the biological aging SVR model, did not have any effect on the respective mean squared errors.